MEDIUM ALGEBRA

# Linear equations in one variable

03

7 questions · ~11 min

SAT QUESTION BANK

2026-05-29

**Q1** ALGEBRA

If $4x - \frac{1}{2} = -5$, what is the value of $8x - 1$?

A. 2

B. $-\frac{9}{8}$

C. $-\frac{5}{2}$

D. $-10$

**Q2**

GRID-IN ALGEBRA

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q3** GRID-IN ALGEBRA

If $\frac{6}{7}p + 18 = 54$, what is the value of $7p$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

$$\frac{1}{4}x + 5 - \frac{1}{3}x + 5 = -7$$

What value of $x$ is the solution to the given equation?

A. -12

B. -5

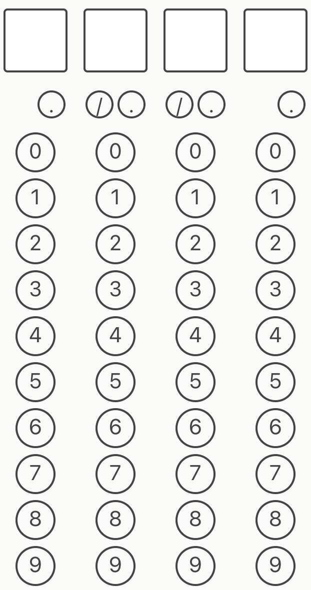
C. 79

D. 204

**Q5** GRID-IN ALGEBRA

An agricultural scientist studying the growth of corn plants recorded the height of a corn plant at the beginning of a study and the height of the plant each day for the next $12$ days. The scientist found that the height of the plant increased by an average of $1.20$ centimeters per day for the $12$ days. If the height of the plant on the last day of the study was $36.8$ centimeters, what was the height, in centimeters, of the corn plant at the beginning of the study?

STUDENT-PRODUCED RESPONSE




Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ALGEBRA

If $9(4 - 3x) + 2 = 8(4 - 3x) + 18$, what is the value of $4 - 3x$?

A. -16

B. -4

C. 4

D. 16

**Q7**

GRID-IN ALGEBRA

A museum rents tablets to visitors. The museum earns revenue of $\$14$ for each tablet rented for the day. On Wednesday, the museum earned $\$406$ in profit from renting tablets after paying daily expenses of $\$112$. How many tablets did the museum rent on Wednesday? $\text{profit} = \text{total revenue} - \text{total expenses}$

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.